

# 01 — HERO

## ALERTEYE

**Stay awake. Get home safe.**

A driver-safety app that detects signs of drowsiness and responds before fatigue becomes a road-safety risk.

**UX Design · Driver Safety · Mobile Interface**

**Team:** Dedeepya N · Sarayu A · Sri Charan · Aryan K · Amrita J

### IMAGE

Use the **AlertEye branding + strongest final UI composition** as the hero. Page 47 is especially useful because it combines the identity with the main driving interface and supporting screens.

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# 02 — BRIEF

**Drowsiness is difficult to manage once it has already taken over.**

The brief was to design a mobile application that uses smartphone technology to detect driver drowsiness and provide timely alerts while driving.

### IMAGE

Use a **driver-at-night contextual photograph** with the phone mounted in the vehicle.

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# 03 — OVERVIEW

**AlertEye turns a phone into a driving companion.**

The system combines visual detection with navigation, alerts, voice assistance and emergency support, creating one continuous safety layer around the driving experience.

## IMAGE

Show the system as:

**CAMERA → DETECTION → ALERT → ACTION → SAFE ARRIVAL**

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# 04 — WHAT DO WE KNOW?

## The existing solutions left gaps in the dark.

The team reviewed existing apps and detection methods and found limitations around low-light performance, response time and detection reliability. This led to exploring camera-based detection alongside an external IR illuminator.

Primary research included **15+ driver interviews** and a **30+ person online survey**, with drowsiness emerging alongside other night-driving concerns such as visibility and animals.

## IMAGE

Use a compact research composition:

**Existing apps → Detection methods → Driver research**

Do not show all the individual research slides.

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# 05 — INSIGHT

## The system needed to notice fatigue before the driver had to react to it.

The research identified observable cues such as **eye closure, blinking, yawning and head movement** as useful indicators of drowsiness. The user journey also showed the importance of escalating the response as fatigue increases.

## IMAGE

Use a simplified detection diagram:

NORMAL → EARLY DROWSINESS → ALERT → ESCALATION

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## 06 — DESIGN RESPONSE

### See better. Respond earlier.

The selected direction combined the phone camera with an **external IR illuminator** to support facial-feature detection in low-light driving conditions. The concept was chosen over relying on the phone camera alone.

#### IMAGE

Use the **phone + IR illuminator setup** from the concept/prototype research.

The actual IR experiment is particularly valuable because it shows the technology being explored rather than simply proposed.

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## 07 — EXPERIENCE

### The alert changes with the driver's state.

The core flow moves from monitoring to detection, then from a gentle warning to a stronger intervention when drowsiness continues. The driver can acknowledge the alert, adjust behaviour, or pull over to rest.

The wider journey continues through navigation, rest-stop suggestions and emergency contact support when required.

#### IMAGE

Use one continuous storyboard:

**START DRIVE → MONITOR → DETECT → ALERT → ACT → SAFE DRIVING →  
DESTINATION**

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## 08 — PRODUCT / UI

## **Safety information stays visible without becoming the distraction.**

The final interface brings together the driving screen, navigation, voice assistance, alert controls, emergency contacts and account settings within a consistent system.

### **IMAGE**

Use a **curated final-screen sequence**, not the full UI board:

**HOME → START DRIVING → MONITORING → ALERT → NAVIGATION**

The final UI uses a dark purple and warm yellow palette across dark and light modes.

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## **09 — DETAILS & VALIDATION**

### **The interaction was built around the moments that matter.**

The interface exploration tested navigation structure, alert behaviour, settings, voice assistance and different visual treatments before converging on the final UI system.

### **IMAGE**

Split the section into:

**UI IDEATIONS → USER FLOW → FINAL UI**

Use the ideation board, flow map and final screens as three visual stages.

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## **10 — ARCHIVE**

### **THE WORK BEHIND ALERTEYE**

A visual archive of the research, detection studies, concept exploration, user flows, interface iterations and technology experiments behind the final system.